Building Infrastructure in LCA - Modelling Approach, Calculation Framework and Default Values

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🏭 **Domain of application:** Textile manufacturing — Knitting (circular, flat, seamless, hosiery, 3D/WHOLEGARMENT), Spinning, Weaving, Cut & Sew, and any capital-intensive industrial process.
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Supporting model: https://docs.google.com/spreadsheets/d/1-jyIQ2yDmoFNVsqEQ8Wwu2TxybaNja9q/edit?gid=714502760#gid=714502760

1. Purpose and Scope

This document describes the methodological framework for including **building infrastructure as a capital good** in life cycle assessment (LCA) models of textile manufacturing processes.

The model was originally developed in the context of **knitting processes** - specifically circular knitting, flat knitting, seamless knitting, hosiery knitting and 3D/WHOLEGARMENT knitting - where it has been validated against machine-specific data and benchmark datasets. However, the underlying logic is general and can be applied without modification to **any capital-intensive industrial process** where floor space is an allocable resource, including:

- Spinning (ring, open-end, air-jet)
- Weaving (shuttle, rapier, airjet, waterjet)
- Cut & Sew and garment assembly
- Dyeing, finishing and washing
- Any other manufacturing process with measurable floor area and throughput

The document is structured as follows: Section 2 establishes the conceptual basis; Section 3 defines the core formula and its parameters; Section 4 presents the default values calibrated for knitting technologies; Section 5 provides guidance for adapting the model to other processes; Section 6 covers implementation in LCA software; and Section 7 lists all supporting references.

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2. Conceptual Basis

2.1 Why include building infrastructure?

Background LCA databases (ecoinvent, GaBi) include construction, maintenance and end-of-life of industrial buildings within specific infrastructure datasets, typically referenced in **m² of gross floor area**. Including these datasets in a foreground process model ensures that the environmental burden of factory construction is attributed — on an amortised basis — to the functional unit of the process.

Omitting building infrastructure systematically underestimates the contribution of capital goods, particularly for **low-throughput processes** where a large floor area is occupied for a relatively small annual output. EF guidelines and ecoinvent methodology both recommend including capital goods and infrastructure when they make a non-negligible contribution to the overall result.

2.2 Allocation principle

Building infrastructure is allocated to the functional unit (1 kg of output) by **amortising the area assigned to a production cell over its total lifetime output**. This approach is consistent with the mass-based allocation of machinery capital goods described in the ecoinvent Guidelines (Weidema et al., 2013) and the EF Guide for compliant datasets (Fazio et al., 2020).

The key insight is that floor area is a finite resource shared across production activities. The fraction attributed to a given machine cell — the **allocated area, A_alloc** — is normalised by the total quantity of product manufactured from that cell over the service life of the building, yielding a value in **m² per kg of output**.

2.3 Background dataset

The building is represented in the foreground model as an input of the ecoinvent dataset:

> 📦 **Dataset:** building, hall / GLO (or equivalent regional variant)
> **Reference flow:** 1 m² of industrial hall
> **Content:** construction, maintenance and decommissioning of the hall
> **Source system:** ecoinvent v3.x

Because the reference flow of the dataset is **1 m²**, the amount entered in the foreground unit process is directly equal to the building intensity in m²/kg — no further conversion is required.

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3. Calculation Framework

3.1 Core formula

The building intensity (m²/kg) is defined as:

$$\text{Building intensity}\left(\frac{\mathrm{m}^2}{\mathrm{kg}}\right) = \frac{A_{\text{alloc}}\ (\mathrm{m}^2)}{Q_{\text{lifetime}}\ (\mathrm{kg})}$$

where Q_{lifetime} is the total mass of product manufactured from the cell over the reference service life:

$$Q_{\text{lifetime}}\ (\mathrm{kg}) = \text{productivity}\ (\mathrm{kg/h}) \times \text{hours/day} \times \text{days/year} \times \text{years}$$

3.2 Parameter definitions

Parameter	Unit	Definition and guidance
A_alloc	m²	Gross floor area allocated to the production cell. Includes machine footprint, access aisles (minimum 0.9–1.2 m on each side), creel or yarn supply area, maintenance clearance, and internal logistics space.

Parameter	Unit	Definition and guidance
		The cell factor (A_alloc / machine envelope) typically ranges from 2× to 8×.
productivity	kg/h	Net output of the machine cell in kg of finished product per operating hour. Taken from OEM specifications or measured at the factory.
hours/day	h/day	Number of operating hours per calendar day. Common values: 8 h (single shift), 16 h (double shift), 24 h (continuous).
days/year	days/yr	Number of production days per year, accounting for planned maintenance shutdowns, seasonal breaks and public holidays. Typical range: 250–330 days/yr.
years	yr	Reference service life of the building or the production cell assignment. Typical range: 10–25 years.
Q_lifetime	kg	Total output over the service life. Derived from the formula above. Serves as the denominator for amortisation.
Building intensity	m^2/kg	The allocated floor area per kilogram of product. This value is entered directly as the amount of the **building, hall** input in the LCA model.

3.3 Cell factor guidance

The **cell factor (CF)** converts the machine envelope area (length × depth from technical drawings) into the allocated production cell area:

$$A_{\text{alloc}} = \text{machine envelope }(\text{m}^2) \times CF$$

Layout type	CF range	Typical conditions
Dense / automated cell	2 – 3×	Automated loading/unloading, narrow aisles, high machine density. Suitable for small repetitive machines (hosiery, circular knitting).
Standard industrial cell	3 – 5×	Standard aisle widths (1.0–1.2 m), operator access on two or three sides, shared creel bay.

Layout type	CF range	Typical conditions
Spacious / maintenance-intensive	5 – 8×	Wide maintenance clearance, offline inspection area, large creel/beam storage, low machine density. Typical for flat or 3D knitting machines with large format.

3.4 Scenario scaling

When the same model is used across multiple operating scenarios (best case, average, worst case), the building intensity scales inversely with lifetime output:

$$\text{Building}_{\text{scenario}} = \text{Building}_{\text{average}} \times \frac{Q_{\text{average}}}{Q_{\text{scenario}}}$$

This is identical in structure to the scaling logic used for **machinery capital goods**, where the denominator is lifetime output and the numerator is a physical quantity attributable to the machine (kg of machine mass for machinery; m² of floor area for buildings).

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4. Default Values - Knitting Technologies

4.1 Adopted defaults

In the absence of site-specific A_{alloc} and Q_{lifetime} data, the following technology-specific reference intensities are applied. They are calibrated from literature benchmarks and cross-checked against machine envelope data from OEM technical documentation.

Process	Technology	m²/kg	Physical plausibility cross-check
Knitting (fabric)	Circular knitting	1.00E-05	Terrot GmbH single-jersey: envelope 5.6 m² × CF 6 → A_alloc ≈ 35 m²; Q_avg ≈ 2.16×10^6 kg → ~1.62E-05 m²/kg [6]

Process	Technology	m²/kg	Physical plausibility cross-check
Knitting (garment/stocking)	Seamless + hosiery	1.61E-05	Santoni SM8-TR1S: 8.5 m² × CF 3.5; Lonati GE616DF3: 1.46 m² × higher CF for multi-machine cell [7,8]
Knitting (panels)	Flat knitting	1.39E-04	KARL MAYER STOLL CMS 503 ki L: 2700×909 mm envelope, lower throughput drives higher intensity [9]
Knitting (3D garment)	3D / WHOLEGARMENT	4.86E-04	SHIMA SEIKI MACH2VS: A=3430 mm, B=3266 mm; lowest throughput → highest intensity [10]

4.2 Technology rationale

Circular knitting operates continuously with minimal interruptions. Its compact cylindrical machine footprint and high throughput yield the lowest building intensity of all knitting technologies.

Seamless knitting and hosiery share a common default because their normalised throughput per allocated cell area is comparable. Seamless machines are physically larger but faster per article; hosiery machines are smaller but deployed in multi-machine cells with shared infrastructure.

Flat knitting (panels) operates with frequent stop-and-start cycles, garment shaping and yarn carrier changes, resulting in markedly lower throughput per m² than circular or seamless technologies.

3D / WHOLEGARMENT knitting produces complete fully-fashioned garments in a single operation, but with complex needle-bed changes and significantly lower knitting speed. This produces the highest building intensity in the group.

4.3 Worked numerical example - circular knitting (average scenario)

Step-by-step derivation for circular knitting:
Machine envelope: 2.55 m × 2.20 m = 5.61 m^2
Cell factor: CF = 6 (creel bay + access aisles + maintenance clearance)
A_alloc: 5.61 × 6 = 33.7 m^2 → rounded to **35 m^2**
Productivity: 30 kg/h (Mayer & Cie. specification)
Hours/day: 16 h (double shift)
Days/year: 300 days
Service life: 15 years
Q_lifetime = 30 × 16 × 300 × 15 = **2,160,000 kg**
Building intensity = 35 / 2,160,000 = **1.62E-05 m^2/kg**
Adopted default: **1.00E-05 m^2/kg** (conservative, same order of magnitude)

The adopted default of 1.00E-05 m^2/kg is slightly below the cross-check result (1.62E-05 m^2/kg) and aligns with the benchmark value reported in the openLCA Nexus organic cotton sweater case study [5]. The difference reflects that the cross-check uses conservative (generous) cell dimensions, while the default targets a median factory layout.

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5. Adapting the Model to Other Processes

The framework described in Sections 2–3 is **process-agnostic**. The only inputs required are: the allocated floor area (A_alloc) and the lifetime output (Q_lifetime). The following guidance covers the most common adaptations.

5.1 Step-by-step procedure for a new process

Step	Action	Guidance
1	Obtain machine envelope	From OEM technical drawing or brochure: length × depth. Use the nominal footprint; exclude moving arms or loading zones that are not permanently occupied.
2	Determine cell factor	Based on plant layout. Use Table in Section 3.3 as a starting point. If a layout drawing is available,

Step	Action	Guidance
		measure directly: A_alloc = total cell area / number of machines.
3	Set productivity	From OEM specification (net output per hour at nominal speed). Adjust for observed efficiency if factory data are available.
4	Set operating regime	Hours/day and days/year from the factory operating model. Use the same scenario as for machine capital goods for consistency.
5	Set service life	Match the building service life to the process assignment horizon. If the machine is expected to be relocated before the building, use the machine lifetime.
6	Calculate Q_lifetime	Apply the formula: productivity × hours/day × days/year × years.
7	Calculate building intensity	A_alloc / Q_lifetime. Enter this value as the amount of **building, hall** in the LCA foreground unit process.
8	Validate	Compare with analogous published values. If the result differs by more than one order of magnitude from comparable processes, review A_alloc and productivity assumptions.

5.2 Indicative ranges for other textile processes

The table below provides **order-of-magnitude reference ranges** based on published LCA data and equipment specifications. These are not validated defaults and should be replaced by process-specific calculations where possible.

Process	Typical productivity	Indicative range (m^2/kg)	Notes
Ring spinning	100–400 kg/h per frame	~1E-06 – 1E-05	High throughput and compact floor; very low building intensity per kg.
Open-end (rotor) spinning	300–800 kg/h	~5E-07 – 5E-06	Highest throughput in spinning; very compact relative to output.

Process	Typical productivity	Indicative range (m^2/kg)	Notes
Shuttle/rapier weaving	20–120 kg/h per loom	~1E-05 – 5E-05	Comparable to circular knitting range.
Airjet weaving	60–200 kg/h per loom	~5E-06 – 2E-05	Higher speed reduces intensity vs. rapier.
Cut & Sew (garment)	3–15 kg/h per workstation	~1E-04 – 5E-04	Labour-intensive; low throughput per m^2 drives higher intensity.
Dyeing / finishing	50–300 kg/h per machine	~1E-05 – 1E-04	Wide range depending on batch vs. continuous process.

⚠️ For informational guidance only. Replace with process-specific calculations whenever possible.

5.3 Multi-process factories

Where a single factory building hosts multiple process types, the total floor area should be apportioned among processes before calculating A_alloc for each. Use one of the following allocation keys:

- **Physical allocation:** measure the floor area actually occupied by each process group *(preferred)*
- **Economic allocation:** apportion by the economic value of output from each process group *(use only if physical allocation is not possible)*
- **Mass allocation:** apportion by the mass of output from each process group *(acceptable for homogeneous product streams)*

6. Implementation in LCA Software

6.1 Generic foreground unit process

In any LCA software that supports ecoinvent as a background database (SimaPro, openLCA, GaBi, Brightway), the building infrastructure input is added as follows:

Field	Entry
Input type	Capital good / infrastructure
Dataset name	building, hall
Geography	GLO (or regional variant if available)
Reference flow unit	m^2
Amount	[building intensity value, m^2/kg] — from Table in Section 4.1 or process-specific calculation
Comment	Amortised building infrastructure; see Building Infrastructure Methodological Document v1.0

6.2 ecoinvent format (unit/kg)

If the LCA model uses the ecoinvent convention of expressing capital goods in `unit/kg`, the amount must be converted:

🔢 Amount (unit/kg) = building intensity (m^2/kg) / 1 m^2 per unit
Since 1 unit of 'building, hall' = 1 m^2, **the numeric value is identical**.
Example (circular knitting): 1.00E-05 m^2/kg → 1.00E-05 unit/kg

6.3 End-of-life

The ecoinvent `building, hall` dataset includes construction, maintenance and decommissioning within a single cradle-to-grave infrastructure dataset. **No separate end-of-life module is needed** in the foreground model. The full lifecycle burden of the building is captured by the single m^2/kg input.

6.4 Supporting files

- https://docs.google.com/spreadsheets/d/1-jyIQ2yDmoFNVsqEQ8Wwu2TxybaNja9q/edit?gid=714502760#gid=714502760

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7. References

7.1 Methodological / rules - capital goods and infrastructure

[1] Fazio, S., Castellani, V., & Sala, S. (2020). *Guide for EF compliant data sets (Version 2.0).* European Commission, Joint Research Centre (JRC). https://eplca.jrc.ec.europa.eu/uploads/JRC_EPLCA_Guide_EF_Data_sets_v2_0-1.pdf

[2] Valente, A., Kusche, O., & Ardente, F. (2022). *Updates on 'Guide for EF compliant data sets (Version 2.0)' to reflect the changes in the Environmental Footprint 3.1 reference package.* European Commission, JRC.

[3] ecoinvent Association. (n.d.). *Infrastructures — construction, maintenance and end-of-life in ecoinvent.* ecoinvent Knowledge Base. Accessed February 2025. https://ecoinvent.org/the-ecoinvent-database/methodology/

[4] Weidema, B. P., Bauer, C., Hischier, R., Mutel, C., Nemecek, T., Reinhard, J., Vadenbo, C. O., & Wernet, G. (2013). *Overview and methodology: Data quality guideline for the ecoinvent database version 3.* ecoinvent Centre. https://ecoinvent.org/wp-content/uploads/2020/10/dataqualityguideline_ecoinvent_3_20130506.pdf

7.2 Benchmark - 'building, hall' as m²/kg in knitting

[5] openLCA Nexus. (n.d.). *LCA case study – Organic cotton sweater (PDF).* Includes: circular knitting with building, hall = 1.61E-05 m^2/kg. Accessed February 2025. https://nexus.openlca.org/

7.3 Machine envelope sources - physical plausibility checks

[6] Terrot GmbH. (n.d.). *Circular knitting machines – Single-jersey (PDF).* Machine dimensions table with frame parameters (e and k). Accessed February 2025. https://www.terrot.de/en/products/

[7] Santoni S.p.A. (2023). *Seamless Technology – SM8-TR1_S (PDF).* Technical drawing: A = 2700 mm, D = 3150 mm. Accessed February 2025. https://www.santoni.com/en/machines/

[8] Lonati S.p.A. (2025). *GE616DF / GE616DF3 / GE516DF / GE516DF3 (ENG brochure PDF).* Dimensions: Width 127.3 cm × Depth 114.8 cm × Height 228 cm. Accessed February 2025. https://www.lonati.com/en/products/

[9] KARL MAYER STOLL Textilmaschinenfabrik GmbH. (n.d.). *CMS 503 ki L machines (PDF).* Machine dimensions: length 2700 mm, depth 909 mm. Accessed February 2025. https://www.karl-mayer.com/en/machines-systems/flat-knitting/

[10] SHIMA SEIKI MFG., LTD. (n.d.). *MACH2®VS WholeGarment® computerized flat knitting machine (PDF).* Dimensions: A = 3430 mm, B = 3266 mm. Accessed February 2025. https://www.shimaseiki.com/product/knit/wholegarment_flat/